

The Gromark cipher is an invention from “The Cryptogram” in 1969 (Hall or DUMBO, Gronsfeld with Mixed Alphabet and Running Key). It’s not a “military” or “spy” cipher. I’d rejected the “Digrafid” cipher as being too contrived and an ACA invention, but there are some points in favour of Gromark here.

Gillogly suggested [Gromark](#) as a possibility for K4 in 1999 to help explain the high index of coincidence at width 50.

OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQJSJSSEKZZWA
TJKLUDIAWINFBNYPVTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR
..*.*.....*.....*.....

There are four coincidences (IC = 0.08) when only $47/26 = 1.8$ would be expected by chance.

(Also, using a [progressive key](#) of length 42, with A, D, G etc “subtracted” we get five coincidences in the first 42, which is the width of a [grid paper](#) in a Sanborn video, but this seems rather contrived. Progressive key usually seems to involve Vigenere after that step – someone would have thought of that ...)

Then the challenge is to find a recurrence relation which has a period of 50 – this would help explain the high index of coincidence at width 50.

One of the simplest ways to do this is to take a 5-digit primer, as in the usual Gromark.

With the primer ABCDE, interpreting the “ $D^{YA}H^R$ ” superscript as a kind of Baconian binary cipher, we can generate the next digit by $A+B^2+C^2+D+E^2$. Then with base 5 or 10, we have a period of 50 in our recurrence relation. E.g. starting with 01033 in base 5, or 51533 in base 10. It is curious that by setting the exponents in the five-digit relation to either 1 or 2 – 32 possibilities - the only way we can get this period of 50 is in this way. It has also been mentioned that the Berlin Clock is related to [base 5](#).

This cipher also fits in well with the “grid paper” method of the original encryption – better than the Hill Cipher or Trifid would, but the method may be too “mathematical” for Sanborn.

After the “BERLINCLOCK” crib came out, it was clear that the default Plaintext row A...Z (e.g. see Blackman’s 1989 Cryptologia paper) would not produce “NYPVTTMZFPK” in the ciphertext with the usual base 10 Gromark. We would need C and R in the plaintext alphabet both producing P in the ciphertext alphabet, but they are more than 10 letters apart in the plaintext alphabet.

The original 1969 article suggested using a keyword mixed alphabet for the cipher alphabet e.g. GRONSFELD giving the ciphertext alphabet DPEKYFJXGAQLMZNHVOCURBTISIW.

The encryption process is still quite prone to error – it may be a good idea to not start with the “BERLINCLOCK” crib when searching.

Nothing has been found yet, but the primers 51533 and 01033 are good places to start (base 10 or 5). For example, starting with 51533 in base 10, several other letters are fixed (B, I, N, O and K). It has been found that this primer produces high-scoring outcomes (out of the 100 possible keys in base 10).

pt **ETXRQ**FLABCDGHIJKMNOPSUVWYZ
ct XFEHSCAYBGUVPJZMNKLRWTQOI

OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSQSSEKZZWATJKLUDIAWINFBNYPVTTMZFPKWGDZXTJCDIGKUHUAUEKCAR
5153337535515979595799331375593591391933339553313351533375355159795957993313755935913919333395533
PAINAUPPLYFORUNDSORTHOMNDSMAKBLIURUISISTDSUZWKHJORNHIMAGOTOSHERBERLINCLOCKSLAIDNPBVIVENTEARAPIERN
.....O...O.....B.....I.L.....K.....N.....

Starting with ciphertext O and Gromark primer digit 5, we find “O” in the ciphertext alphabet and go five places left, which leads to “P” in the plaintext alphabet, and so on.

Naturally, this hasn’t led to any exciting plaintext discoveries...

.515333 .OBKRUO
7535515 XOGHULB
9795957 SOLIFBB
9933137 WFLRVQQ
5593591 PRNGKSS
3919333 OTWTQSJ
3955331 QSSEKZZ
3351533 WATJKLU
3753551 DIAWINF
5979595 BNYPVTT
7993313 MZFPKWG
7559359 DKZXTJC
1391933 DIGKUHU
3395533 AUEKCAR